Original RTL: 8-bit SAD Saturation

VCS + Verdi | Directed high-contrast test

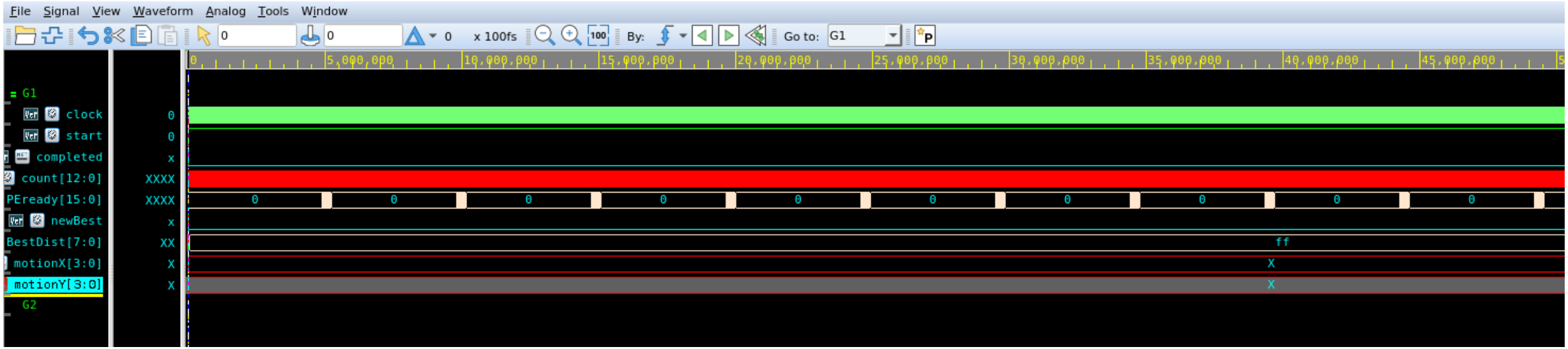
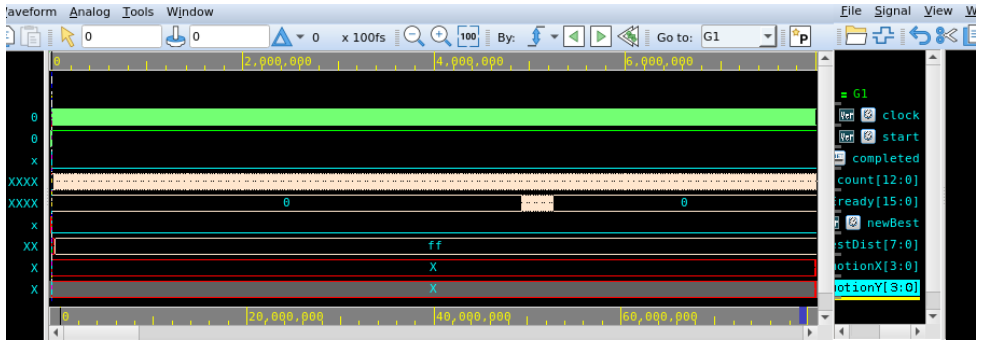


Figure 1. PEready updates while newBest stays low; BestDist remains FF.



At completion:
BestDist = FF
motionX = X, motionY = X

All 256 pixels differ by FF, so the correct SAD is FF00. The 8-bit comparator cannot select an FF candidate against an FF reset value.

16-bit SAD: Corrected Motion Vector

VCS + Verdi | Separate RTL and self-checking testbench

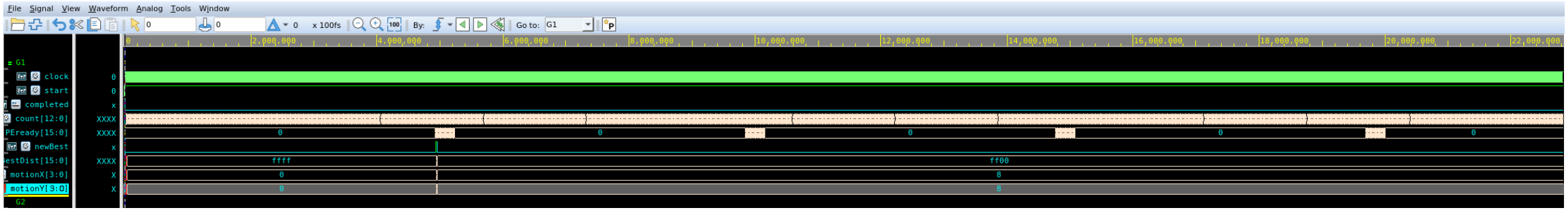


Figure 2. newBest asserts and BestDist updates from FFFF to FF00.

Original RTL

BestDist = FF
motionX / motionY = X / X

16-bit corrected RTL

BestDist = FF00
motionX / motionY = 8 / 8

Encoded 4-bit vector (8,8) is signed (-8,-8). With equal SADs, the first candidate wins; the completion-hold test kept outputs stable.

14 nm Synthesis: Area and Timing

Fusion Compiler | SAED14 RVT | 520 MHz

Mapped design comparison

Metric	Original 8-bit	16-bit V3B
Standard-cell area	881.96 µm²	1,193.03 µm²
Mapped cells	1,693	1,905
Sequential cells	280	416
Memory macros	0	0

Timing constraints

Clock: 1.923 ns (520 MHz)
Input delay: 0.5 ns
Output delay: 0.5 ns
Clock uncertainty: 0.2 ns

Compile-stage setup slack

+0.06 ns (MET)

The 16-bit datapath uses 311.07 µm² more cell area (+35.3%) to preserve the full SAD. Synthesis, floorplan and placement are recorded; final routed setup/hold are separate checks.

Reference and search memories are in the testbench, not in the synthesized motion estimator.